Estimated Position of the Moving Axes of a Mechanical Arm from an Unknown State

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Introduction

- **Context:** Computer vision drives essential advances in robotics and industrial automation.
- **Problem:** Traditional robot manipulator pose estimation methods – such as RoboPose and CtRNet-X – rely on prior joint angle state readings, limiting their applicability when these data are unavailable.
- **Challenge:** Recent holistic models, such as HRPE (Holistic Robot Pose Estimation), estimate pose from a single RGB image but require massive volumes of synthetic data, increasing training time and computational cost.
- **Objective:** Estimate the 3D pose and perform segmentation of a 7-DOF (Franka Emika Panda) articulated robotic manipulator from a single RGB image, without prior knowledge of joint states.

Contributions

- **Depth Feature Fusion:** Integration of the Depth Anything V2 monocular depth extractor into the HRPE architecture without modifying its original structure.
- **Sim-to-Real Self-Supervised Training:** A two-stage strategy combining synthetic pretraining and self-supervised refinement on real data, guided by kinematic and segmentation consistency constraints.
- **Data Efficiency:** The synthetic dataset was reduced from >26,000 to only 8,000 images (~70% reduction), outperforming the baseline model.

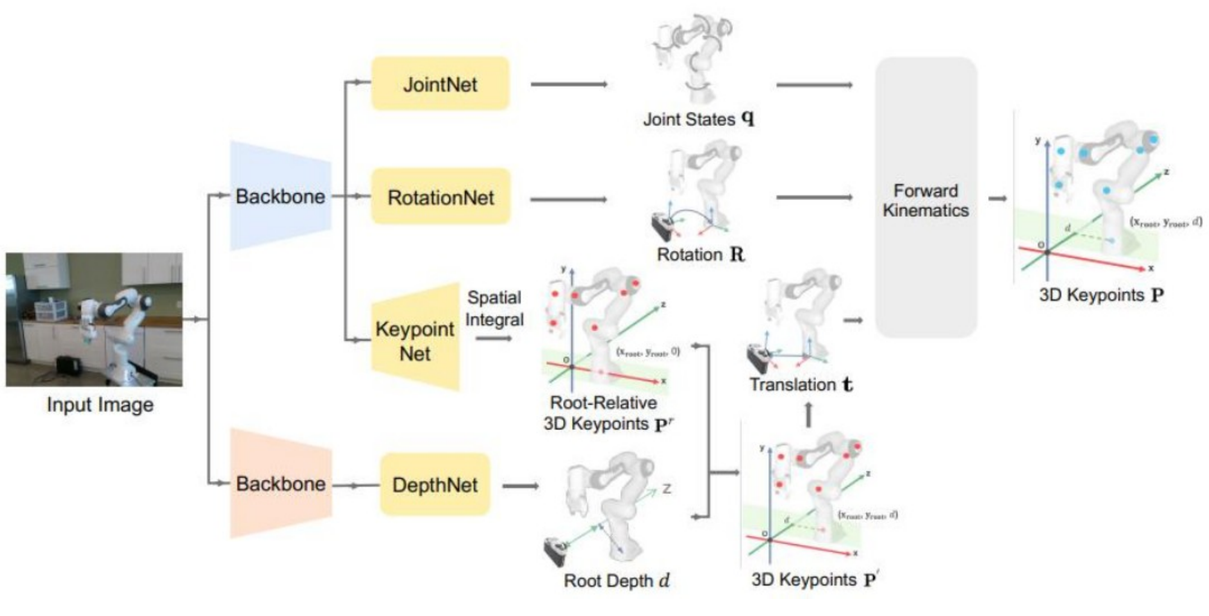


Figure 1. General structure of the Holistic Robot Pose Estimation (HRPE) model. Adapted from (Ban, 2024).

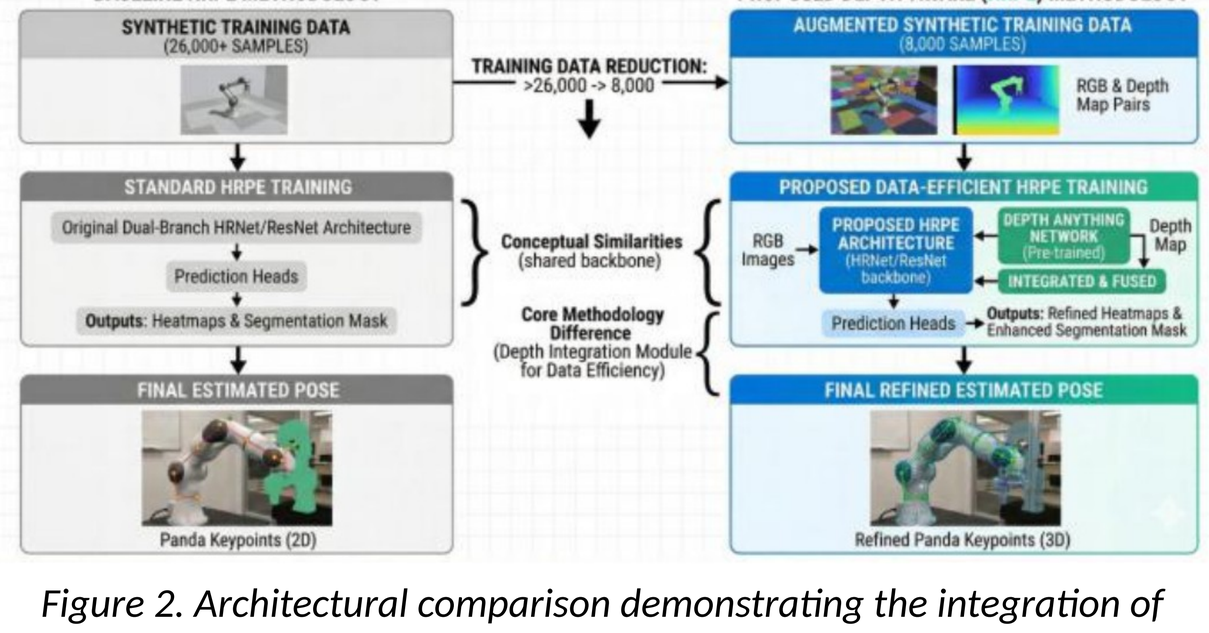


Figure 2. Architectural comparison demonstrating the integration of depth information by Depth Anything V2 into HRPE.

Methodology

Camera Model e Geometric Foundations: 3D point mapping onto the image plane via a pinhole model, defining the Network Architecture:

- **Depth Anything V2 Module:** Extracts depth maps and heatmaps based on DINOv2 and DPTHead. Features are concatenated, compressed, and fused into the HRNet-32 branch.
- **Parallel Extractors (HRPE):**
  - **DepthNet** (HRNet-32), **JointNet**, **RotationNet**, and **KeypointNet** (shared ResNet-50).
- **Forward Kinematics (FK):** Combines **q**, **R**, and translation vector **t** to compute final 3D positions in **P**.

Training Strategy (Adam Optimizer):

- **Stage 1 (DepthNet Pretraining):** Synthetic data, 500 epochs, learning rate (LR) 10^{-4}, batch size 32.
- **Stage 2 (Full Model Synthetic):** 8,000 synthetic images, 400 epochs, LR 10^{-4}, batch size 16.
- **Stage 3 (Self-Supervised Real-Data Fine-Tuning): 400 epochs, LR:** 400 épocas, LR de 10^{-5}, batch size 16.

Method	Synthetic Samples	Accuracy (%)
RoboPose (Labbé, 2021)	–	74.3
HRPE (Ban, 2024)	>26,000	75.2
Proposed method	8,000	75.4

Table. 1. Comparison with state-of-the-art methods on the DREAM RealSense dataset.

Results

Efficiency Analysis & Evaluation on the DREAM Benchmark (RealSense Split): Evaluated using the Area Under the Average Distance Curve (AUC/ADD) metric at a 60 mm threshold:

- **70% Computational Reduction** in training iterations per epoch during the synthetic phase.
- **Depth Anything V2 Overhead** restricted strictly to feature extraction.
- **Learning Curves** showing stable convergence without severe overfitting in the self-supervised phase.

Conclusions

- **Depth Feature Fusion** increased the model's geometric representation richness without modifying the basic HRPE architecture.
- **Self-Supervised Real-World Training** ensured strong generalization capability.
- **Significant Synthetic Data Reduction** substantially decreased computational cost while maintaining high performance.
- **Future Work:** Validation on robotic platforms, evaluation of complementary 2D metrics, calculation of per-joint angular error, and mitigation of dependency on segmentation masks and URDF models.